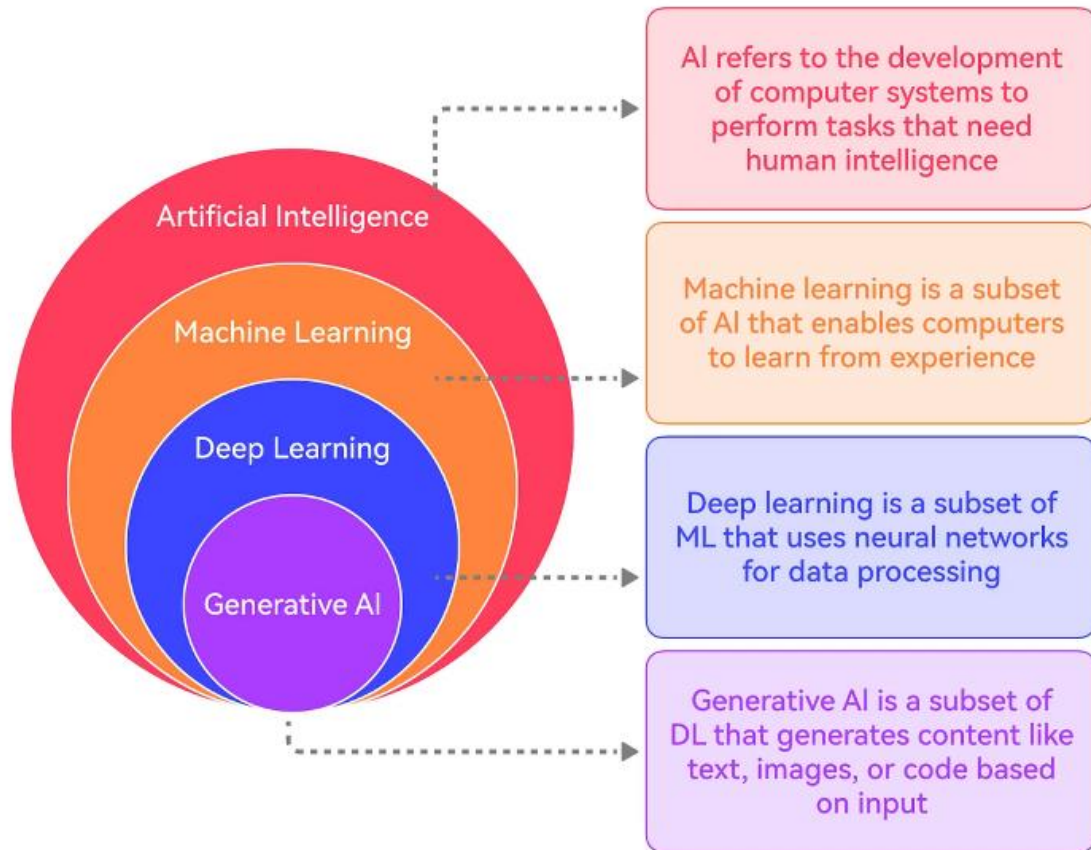


Understanding the GenAI Terminologies

One of the biggest obstacles to getting started with GenAI is not understanding the basic terminologies.

Let's cover the most important things to know about.



Generative AI:

- AI application which generates Text, Image, Audio, Video

How will It generate?

- is already trained on vast data using the concept of neuro networks (how human think) using intelligence algorithm (ML, DL) by some Big Organizations
- Pre-Trained models (somebody already trained a model)
- We are using those trained model, is called transfer learning.

Generally, Models are of two types

- Already trained by some other big organizations (Pre-Trained Models)
- We train the Model with our data sets (Custom Models)

Generative AI

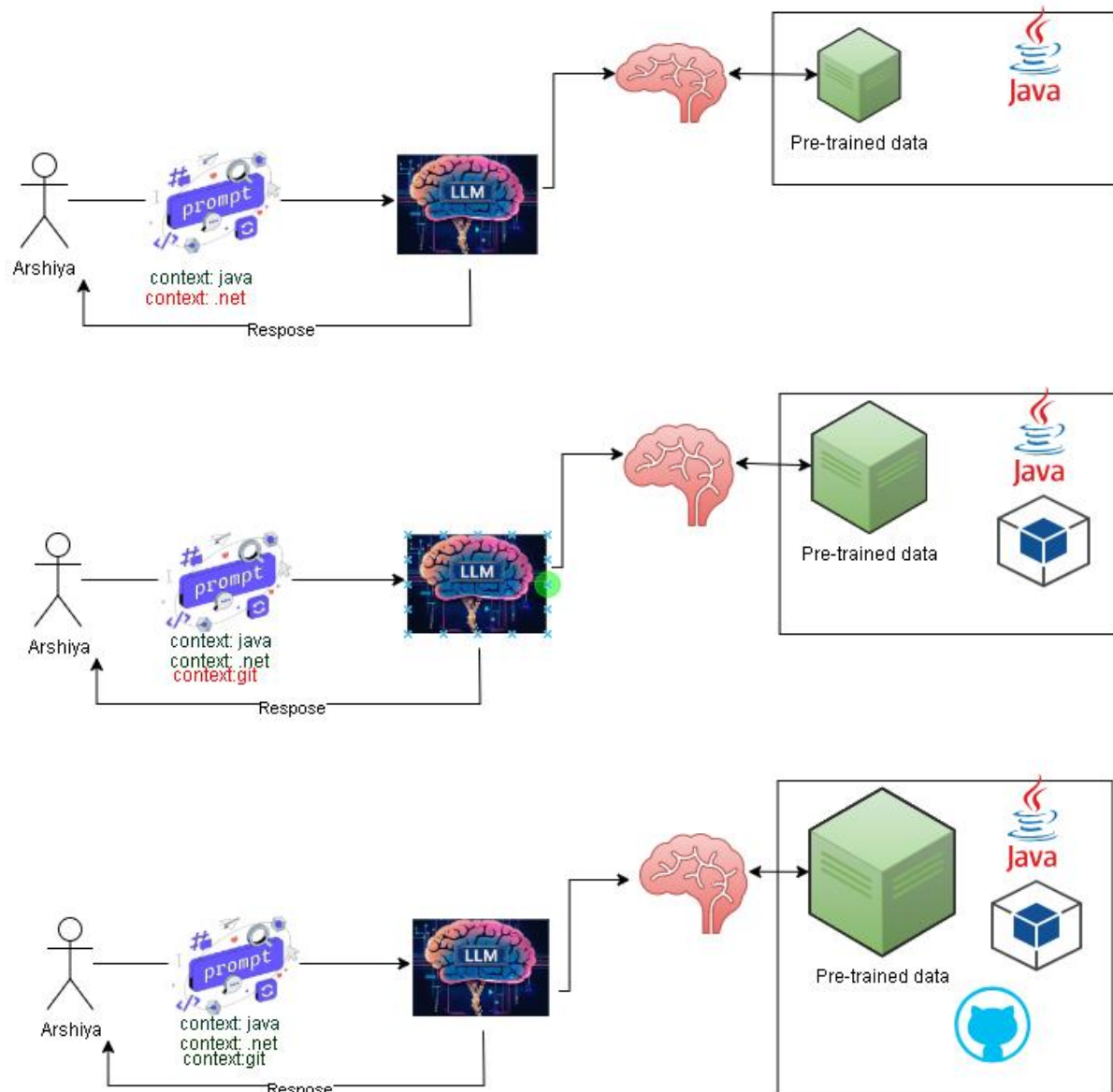
In Market there are already many companies available with their own LLMs. some of them are

- OpenAI
 - ChatGPT (Generative Pretrained Transformers)
 - Sora # video from text
- Google
 - Gemini Models
- Meta (Facebook)
 - Llama
- Mistral AI
 - Mistral 7B
- Amazon
 - Bedrock Models
- Anthropic
 - Claude
- xAI
 - Grok

How to Use GenAI?

- Connect
 - If we want to access Facebook, Gmail we required our password, same way If want to access GenAI we need **API Key**.
- Communicate
 - We need to talk with models, so we need to use some communication mechanism -> Prompt
 - While we are talking with GenAI models, we may give some examples.
 - Without example- Zero Shot Prompt
 - With one example- One Shot Prompt
 - With more example- Multi Shot Prompt

Limitation of Generative AI (Reinforcement Learning)?



- Exponentially Increase Data Store (like **monolithic application**)
 - In a microservice architecture, each service is specialized for a specific function, ensuring scalability and flexibility.
- Information only to topics in which its trained, no real-time information
- Provide Information and Advice
- Can't take decision or any action

What Next?

Generative AI (LLM) is already a trained model, and solution of its limitations

Human Capabilities:

Data Collection: A human can gather relevant data from various sources.

Situation Understanding: A human can comprehend the context and analyze the situation at hand.

Decision Making: A human can evaluate the information and make informed decisions.

Action Execution: A human can perform actions based on the decisions made, executing tasks

Machine Capabilities (Agentic AI):

In contrast, if a machine is equipped with the ability to perform the same functions, the following capabilities are exhibited:

Data Collection: The machine can gather data from relevant sources in real-time, just as a human would.

Situation Understanding: The machine can analyze the situation, interpreting the context and processing the gathered data effectively.

Decision Making: The machine can assess the data, apply predefined rules or learned models, and make decisions autonomously.

Action Execution: The machine can carry out tasks or actions based on its decisions, automating the process with precision and efficiency.

This combination of capabilities, where a machine can replicate human-like cognitive functions such as understanding, decision-making, and execution of tasks, is referred to as **Agentic AI**.

Agentic AI: Understanding the Next Evaluation of AI

As a specialized agent—whether in travel, finance, or any other field—here’s how a trained model can operate effectively with external resources to deliver optimal service:

Expert Knowledge Base: The model is trained specifically for a particular domain, gaining deep insights and expertise. For example, a travel agent model understands

destinations, booking systems, and itineraries; a financial agent model knows investment options, market trends, and financial products.

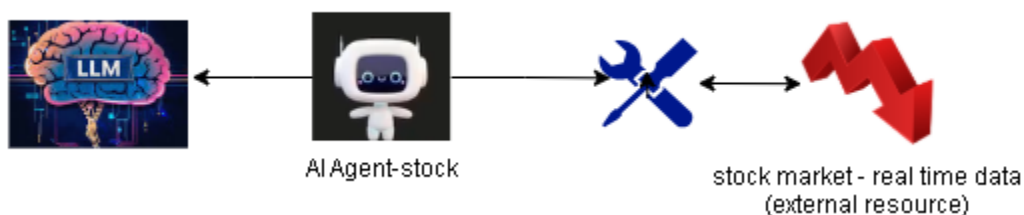
Access to Relevant Resources: Just like an experienced agent who knows where to look for information, the model can access a variety of external resources:

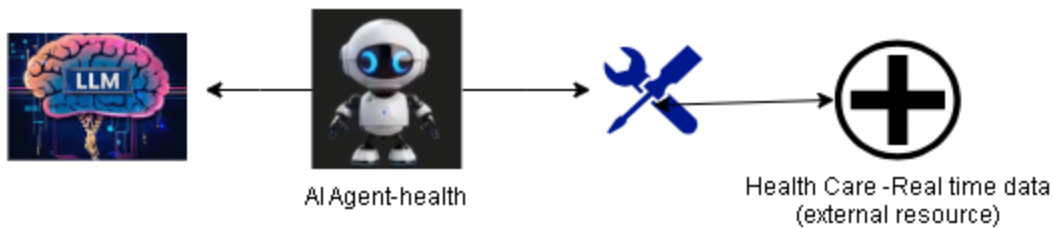
- **Real-Time Data:** Whether it's flight availability, current stock prices, or changing market conditions, the model can pull in the latest data for accurate and timely recommendations.
- **Artifacts (API Endpoints):** The model can interact with APIs, databases, and external systems, retrieving necessary information—like hotel availability, insurance rates, or tax regulations—to provide relevant advice.
- **Other Agents:** Collaboration with other specialized agents or models can expand the scope of service. For example, a financial model might query a market analysis agent, or a travel agent might collaborate with a weather data model.

Continuous Learning: As the model interacts with these resources and gains new insights, it evolves:

- **Real-Time Knowledge Updates:** The model's knowledge base gets updated automatically with fresh information from real-time data and artifacts, ensuring recommendations stay accurate and relevant.
- **Self-Improvement:** By accessing and learning from external data sources, the model continuously refines its expertise in the domain, just like an expert who keeps up to date with the latest trends and resources.

Efficient Decision-Making: With access to all these resources, the model becomes an expert at extracting the most relevant data, analyzing it, and making well-informed decisions—just like a professional who knows exactly where to source the right information quickly.





How Does Agentic AI Works?

- **Perception**
 - AI gathers data from sensors, databases, or real-time sources, e.g., ML, DL, NLP, AI all needs data (you will give me data === I will provide the insight)
 - agents need real time data
- **Planning**
 - Process information and formulates strategies.
 - e.g. when we go to travel agent his/her fields know how system works, they can guide us very well step by step process- planning
- **Action**
 - Executes decisions without direct human commands
- **Learning**
 - Continuously updates its model based on outcomes

Key Features of Agentic AI

- **Autonomy:**
 - Operates without human intervention.
- **Goal Oriented**
 - *Works towards achieving predefined objectives*
- **Adaptability**
 - Learns from new experiences
- **Decision-Making**
 - Makes real-time decision based on data
- **Self-Learning**
 - Improves performance over time

Applications of Agentic AI

- Autonomous Vehicles
 - self-driving cars and drones (sense traffic, make driving decisions)
- Health Care
 - AI-driven diagnostics and robotic surgeries
 - *superbug Mistry*: <https://www.ndtv.com/world-news/ai-solves-years-long-superbug-mystery-in-48-hours-shocks-scientists-7761512>
- Stock Trading
 - Analysis market, execute trades
- Cyber Security
 - AI-driven threat detection
- Smart home assistant
 - Adjust AC, turn on light

Future of agentic AI

neogama: <https://www.youtube.com/watch?v=uVcBa6NXAbk>

Components of AI Agents

- Perception Module
 - Gathers inputs (text, images, voice, videos etc.)
- Memory
 - Stores past interactions for context
- Reasoning Engine
 - Use AI Models to analyze and infer
- Decision-Making System
 - Determines Optimal actions
- Execution Modules
 - Acts upon decisions (API calls, commands etc.)
- Learning Mechanism
 - Improve performance with time.

Why Do AI agents required External tools and LLMs?

Any AI Agent requires two main things

- Real Data
- Any Generative AI LLM Model
 - LLM provides reasoning, comprehension, and language skills
 - No real data

How to get real time data?

Enhanced Capability through Tools

- Access to real world data
 - **Integrate APIs**, databases and Knowledge sources
- Task Automation
 - Uses **external tools** like search engines, calculators and APIs.
- Better Accuracy and Efficiency
 - Combine AI reasoning with structured tools

Use Case: Financial Advisor Agents

Role: Provides Investment suggestions, tax planning and risk analysis.

Data Collection Agent

- Collects financial data from users

Financial Analyst Agent

- Analysis market trends and portfolio performance

Strategy Suggestions Agent

- Suggests optimal investment strategies

Client Interaction

- Uses LLM for client interactions and reports

Think: Complete flow requires ***multi agents***,

Multi-Agent Systems (MAS) in Action:

In many complex scenarios, a single agent may not be sufficient to handle the entire flow of tasks. Instead, multiple agents working together are required to efficiently process and manage tasks across various stages. This is where a Multi-Agent System (MAS) comes into play.

- ***Is it possible to use only one agent and will give all the work?***
 - Monolith Vs Microservice

- robot-commits-suicide:
 - <https://www.indiatoday.in/technology/news/story/robot-commits-suicide-in-south-korea-because-he-was-made-to-do-a-lot-of-work-2562775-2024-07-05>

A Multi-Agent System involves a network of specialized agents, each designed to handle specific tasks or components of a process. These agents collaborate, share information, and coordinate actions to achieve a common goal. The system is particularly effective in dynamic and complex environments where single-agent solutions may fall short.

Why Use a Multi-Agent System?

Distributed Problem Solving: MAS allows tasks to be divided across multiple agents, enabling better scalability and efficiency.

Collaboration: Agents can communicate and work together, leveraging their individual strengths to tackle complex problems.

Flexibility: MAS can adapt to changing conditions, with agents able to adjust their behavior based on new data or shifting objectives.

In summary, **Multi-Agent Systems** offer a powerful approach to solving complex problems by integrating multiple intelligent agents that can collaborate, share information, and perform tasks autonomously.

Popular Framework: The MAS

Crew AI	AI Agent framework for task collaboration.
LangChain	Helps in building AI-powered applications

Popular AI agent tools

Google Search API	Fetches real-time search results directly from Google, allowing AI agents to gather up-to-date and accurate web data.
DuckDuckGo API	A privacy-focused search engine API that provides instant answers while ensuring user anonymity.
SerpAPI	Automates the scraping of search engine results from Google and other search engines, providing structured data.
Serper API	A powerful tool for extracting search results, designed specifically to interface with Google Search and provide clean, structured data.

These **search tools** for AI agents enable access to a wide range of information sources, from real-time web results and privacy-focused queries to computational intelligence and structured insights. By utilizing these APIs, AI agents can efficiently gather and process relevant data, making them more capable of providing timely and accurate responses in various domains.

Environment:

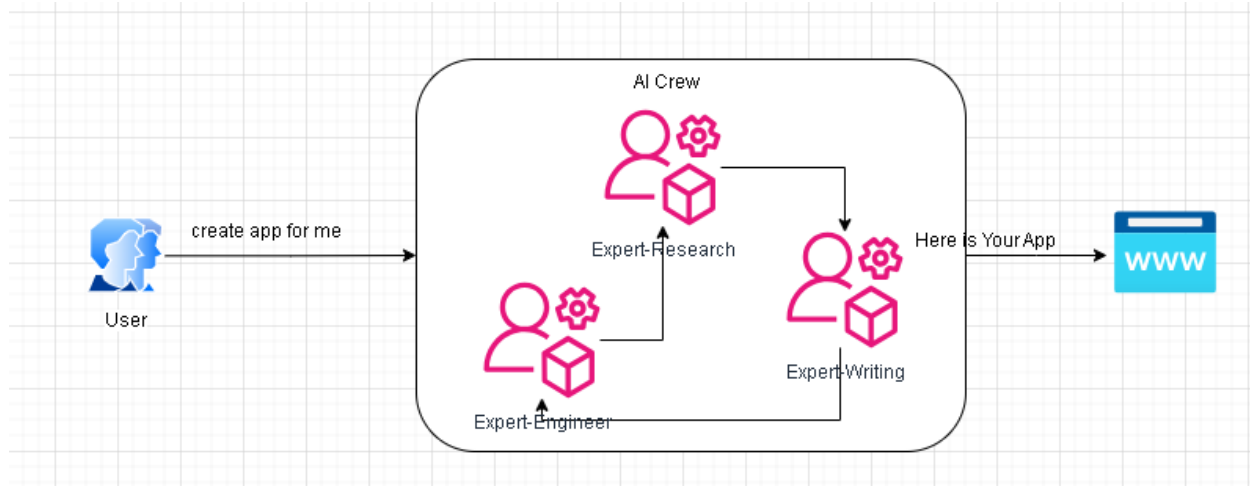
IDE	Google Colab, Cursor AI	
Framework	Crew AI/ LangChain	
LLM	Google Gemini,llama	API Key
Agent Tool	Serper Tool/SerpAPI, DuckDuckGo	API Key
Coding Language	Python	
Development Tools	anaconda, streamlit, pyngrok	API Key -ngrok

CrewAI (A complete framework for building and orchestrating multi-agent AI systems)

Crew AI is a framework that lets you create teams of AI agents (like virtual workers) to solve complex tasks together. Think of it as building a "crew" where each member has a specific role, and they collaborate to get things done.

CrewAI's main components:

- **Agents:**
 - Individual AI workers with specific skills (e.g., a "researcher" agent, a "writer" agent).
 - Each agent has a role (e.g., "Customer Support Agent") and tools (e.g., access to external resource), you can use any existing ones from LangChain or serper or others.
- **Tasks:**
 - Jobs (small and focused mission) assigned to agents (e.g., "Answer a customer complaint" or "Write a report").
- **Process (Orchestrator):** This is the workflow or strategy the crew follows to complete tasks.
 - Which agent does what.
 - When to pass work from one agent to another.
 - How to handle errors or retries.
- **Crew:** Where Agents, Tasks and a Process meet, this is the container layer where the work happens



Why Use Crew AI?

- Efficiency: Agents split work (like a real team).
- Scalability: Add more agents for bigger tasks.
- Flexibility: Swap tools or roles as needed.

Agent

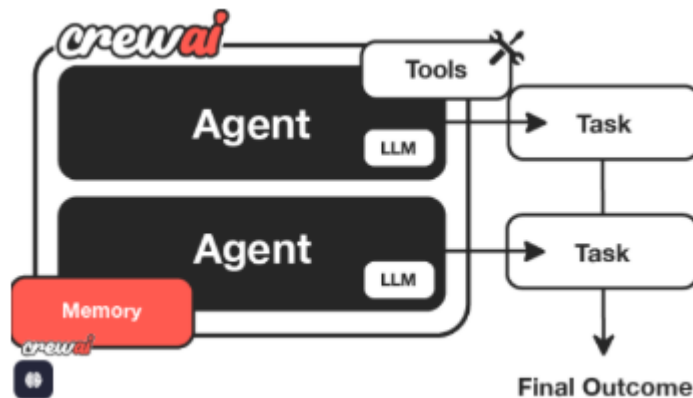


Attribute	Description
role	The role defines their expertise
goal	The goal defines the exact task they need to finish
backstory	The backstory of the agent gives more context about the role and goal.
tools	Helper to access external resource
llm	Specific llm
verbose	gives more output to, what agent is doing behind the scenes in the terminal
memory	Whether agent should have memory or not
allow_code_execution	enable the code execution for the agent



Task

description	
expected_output	
agent	
output_file	



Parameters	Description
tasks	list of tasks assigned to crew
agents	list of agents which are part of the crew
process	The process flow that the crew will follow (e.g. sequential)
verbose	The verbosity level for logging during execution, Defaults to false
manager_agent	sets a custom agent that will be used as a manager
planning	Adds planning ability to the crew. When activated before each before each crew iteration. All the crew data is sent to an Agent Planner that will plan the tasks, and this plan will be added to each task description